

* LADE Extra Assignment *

Q1] Application of Linear Transformation & its examples
 → Linear Transformation is a function between two vector spaces that preserves the operations of vector addition & scalar multiplication.

Applications:-

- 1) Computer Graphics - Used to rotate, scale or move objects in 2D & 3D space.
- 2) Data Compression - Image & audio compression.
- 3) Machine Learning - Used in Principal Component Analysis for dimensionality reduction.
- 4) Robotics - Coordinate transformation for robot arms & sensor mapping.

Example:-

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined as

$$T(x, y) = (2x + y, x - y)$$

It satisfies both additivity & homogeneity.

Q2] Application of Eigen value Eigen vectors & its examples
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Given a square matrix A , a non-zero vector v is an eigen vector if $Av = \lambda v$, where λ is corresponding eigenvalue.

Applications:-

- 1) Stability Analysis - Used in control systems to determine stability of dynamic systems.
- 2) Quantum Mechanism - Observable are represented by operators whose eigenvalues represent measurable quantities.
- 3) Facial ~~face~~ recognition - Eigenfaces technique reduces image dimensions using eigenvectors.

4) Google Page Rank - Determines websites ranking using eigenvector-based algorithms.

Example:-

For matrix, $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

Find eigenvalues by solving

$$\det(A - \lambda I) = 0$$

$$\Rightarrow (4 - \lambda)(3 - \lambda) - 2 = 0$$

$$\Rightarrow \lambda^2 - 7\lambda + 10 = 0$$

$$\therefore \lambda = 5 \quad \& \quad \lambda = 2$$

$$= \quad =$$

These are eigen values. Corresponding eigenvectors can then be computed.

Q3] Application of Differential Equations & examples.

Differential equation is an equation that involves a function & its derivatives.

Applications:-

- 1) Physics - Describes motion, heat, waves & electricity
- 2) Biology - Models population growth & spread of disease.
- 3) Engineering - Used in control systems, electrical circuits & fluid mechanics.
- 4) Economics - Models economic growth, interest rates & investment.

Example :-

simple population growth $\Rightarrow$

$$\frac{dp}{dt} = rP \quad - [P = \text{population} \ \& \ r = \text{growth rate}]$$

$$\therefore P(t) = P_0 e^{rt}$$

This represents exponential population growth over the time.